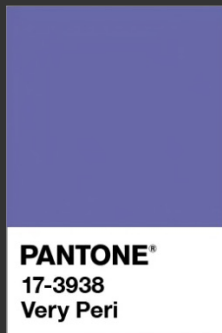
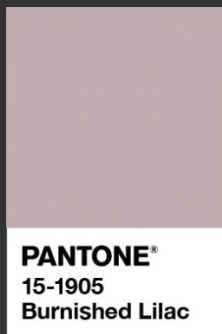


hexadecimal representation

CSCI
134

pantone is
a company
that names
colors



“

As we move into a world of unprecedented change, the selection of PANTONE 17-3938 Very Peri brings a novel perspective and vision of the trusted and beloved blue color family, encompassing the same time with its violet red undertone, PANTONE 17-3938 Very Peri displays a spritely, joyous attitude and dynamic presence that encourages courageous creativity and imaginative expressions.

— LEATRICE EISEMAN, EXECUTIVE DIRECTOR
OF THE PANTONE COLOR INSTITUTE

”

ANNOUNCING THE PANTONE COLOR OF THE YEAR 2022

PANTONE 17-3938 Very Peri

A New Pantone Color Whose Courageous Presence Encourages Personal Inventiveness and Creativity.



a common way that
computers represent

color

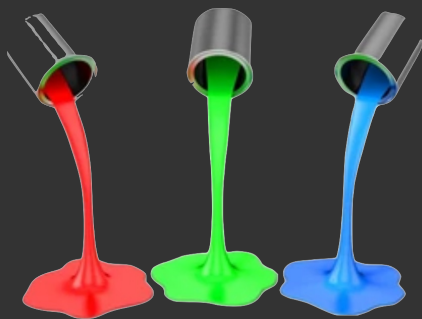
is an **rgb** encoding



0-255
units of
red

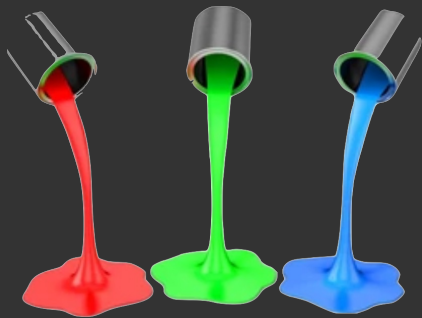
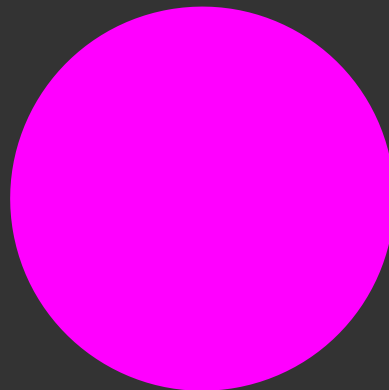
0-255
units of
green

0-255
units of
blue



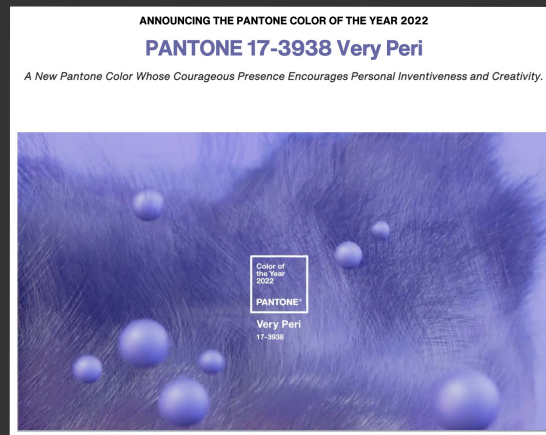
255 0 255

=



102 103 171

=



why do you think
these range from
0 to **255**?



0-255
units of
red

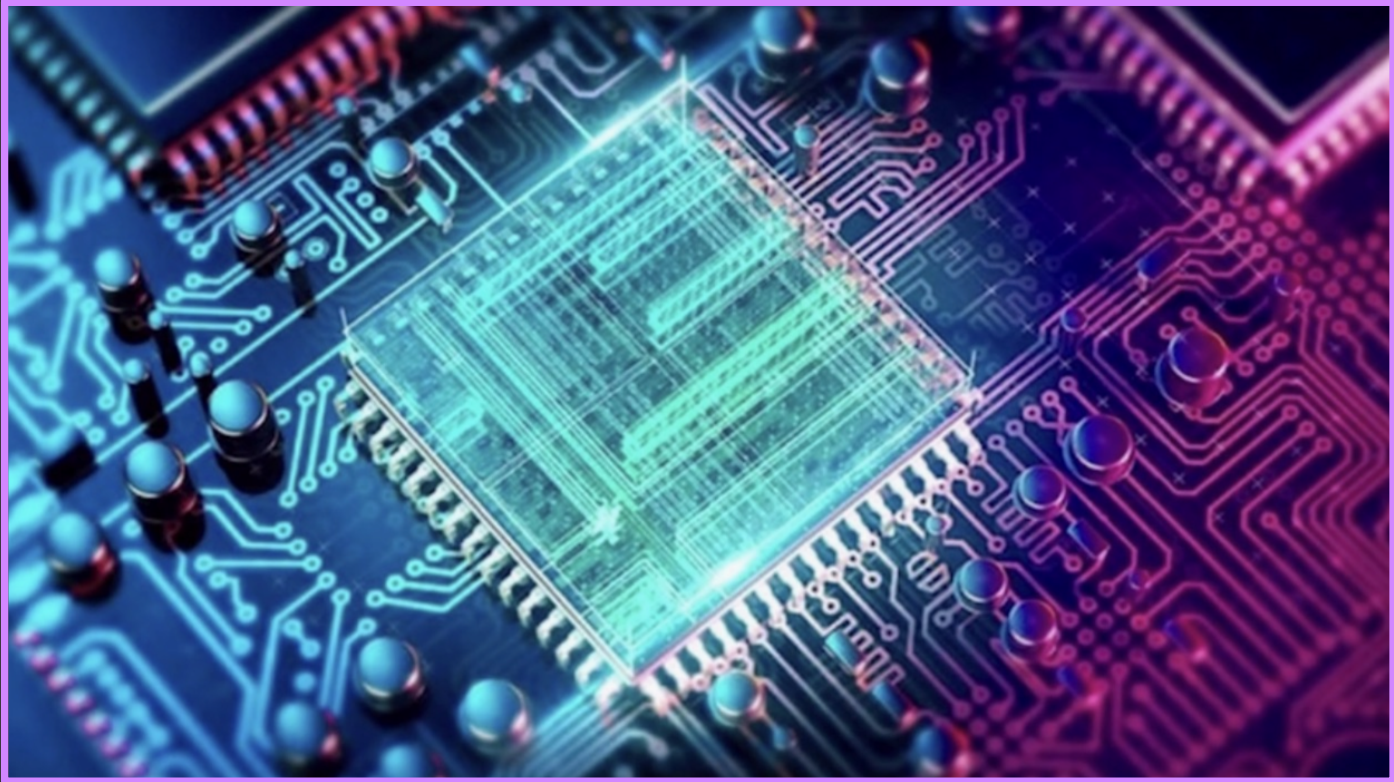
0-255
units of
green

0-255
units of
blue

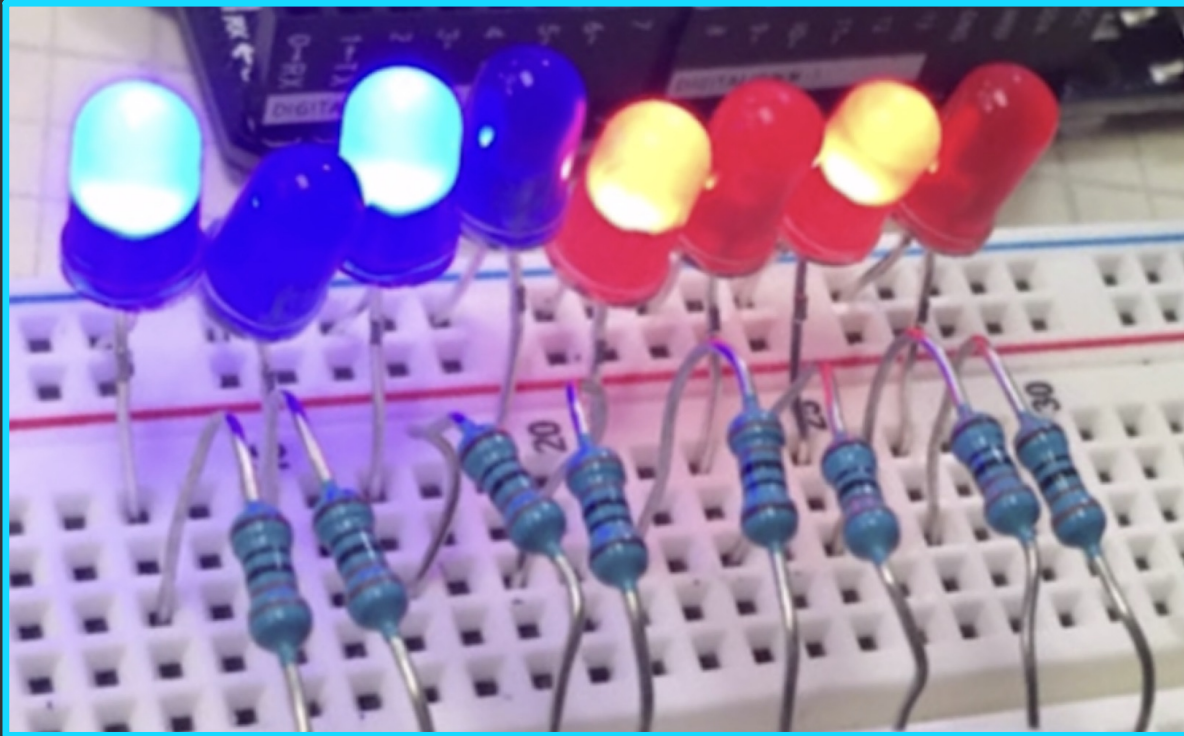
if we look deep
inside a computer



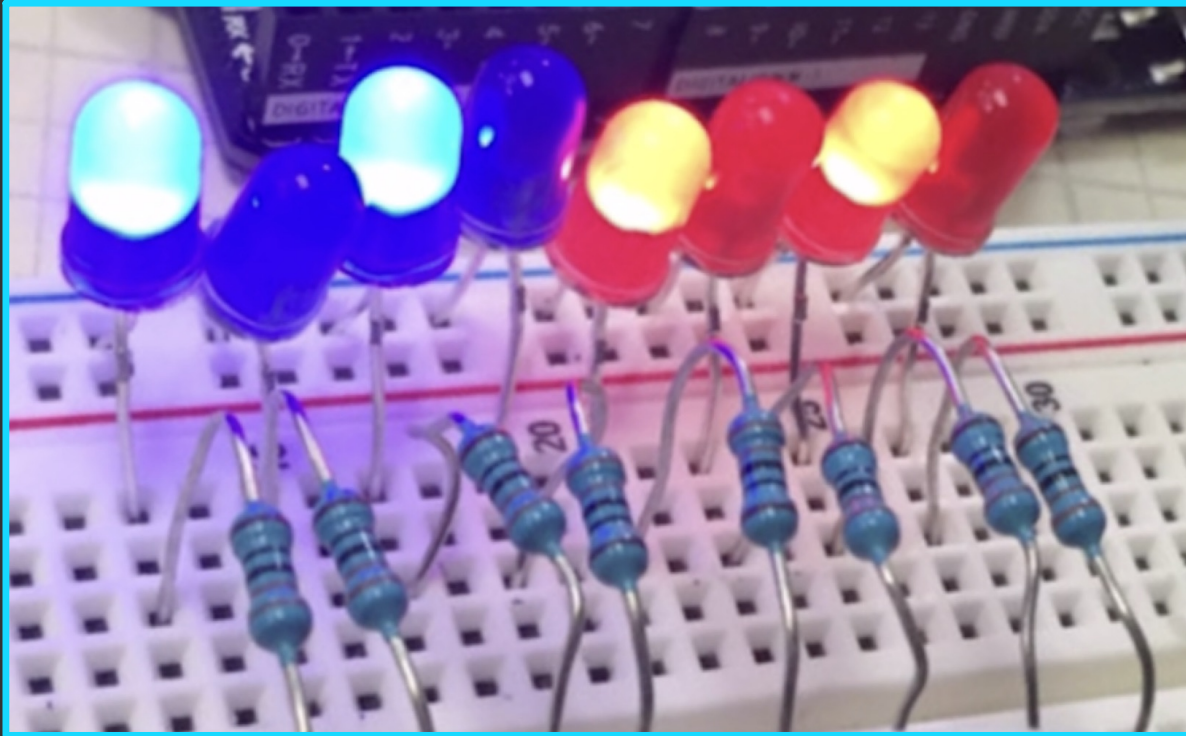
there are lots of tiny "lights"



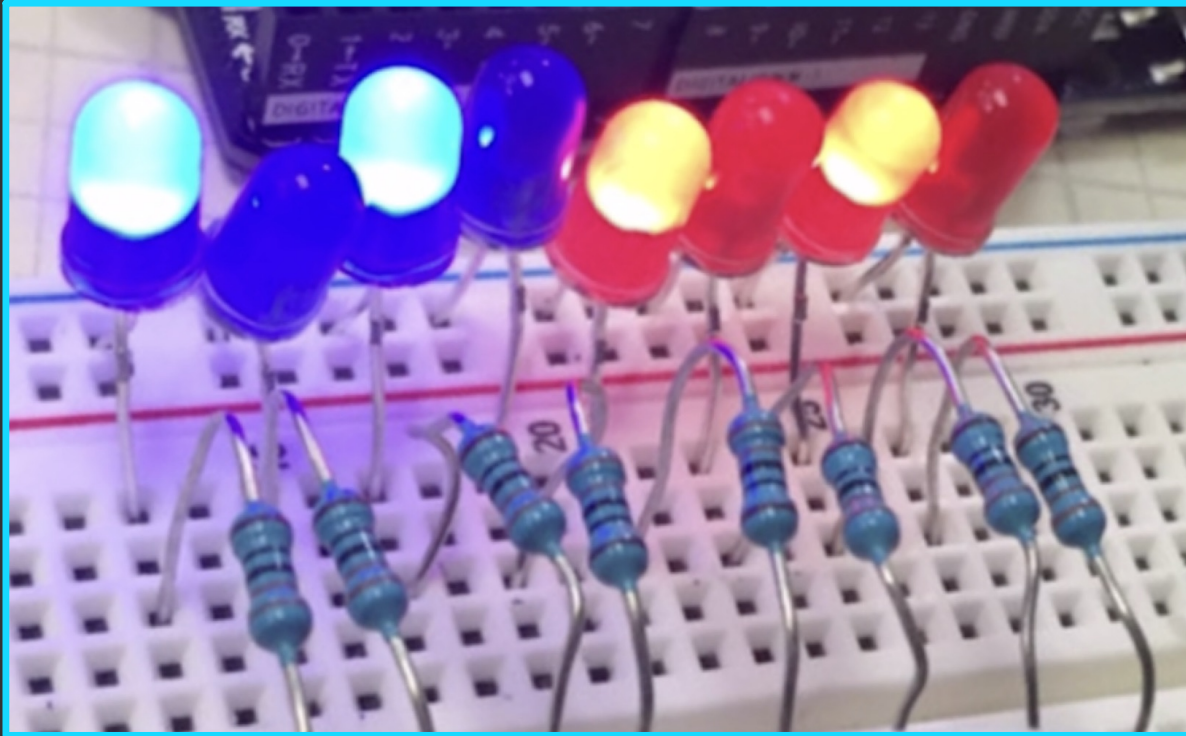
arranged in groups of 8



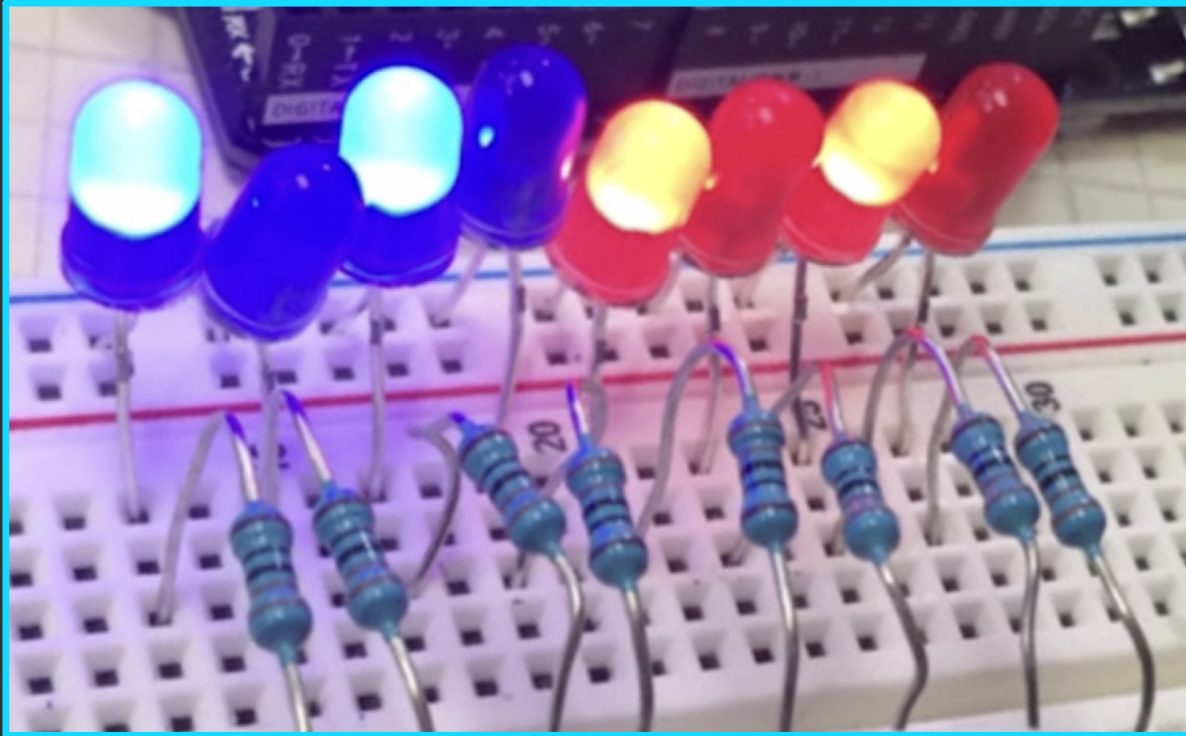
how many values can be encoded by 8 lights?



$2^8 = 256$ values can be encoded by 8 lights



each group of 8 lights is called a **byte**



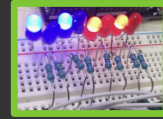
since each byte can
represent **256** values,
we can store a
color
using **3** bytes



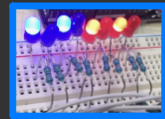
0-255
units of
red



0-255
units of
green



0-255
units of
blue



like this



102

103

171

2022 Pantone
Color of the Year



PANTONE
17-3938 TCX
Very Peri

Color Hex Code
#6667AB



but what's
this?



102

103

171

2022 Pantone
Color of the Year



PANTONE
17-3938 TCX
Very Peri

Color Hex Code
#6667AB



half a byte
is called a
nibble



102

103

171

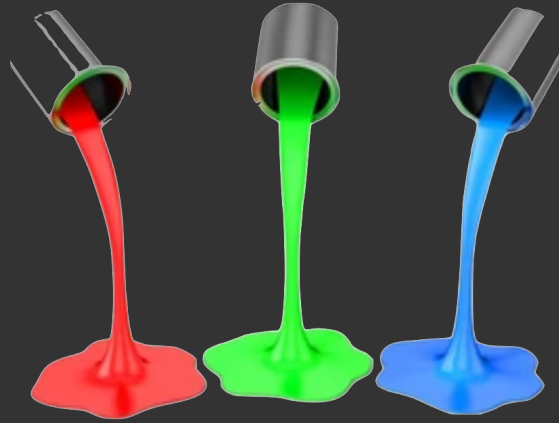
2022 Pantone
Color of the Year



Color Hex Code
#6667AB



since a **nibble**
is **4** bits, it
can represent
 $2^4 = 16$ values



102

103

171

2022 Pantone
Color of the Year



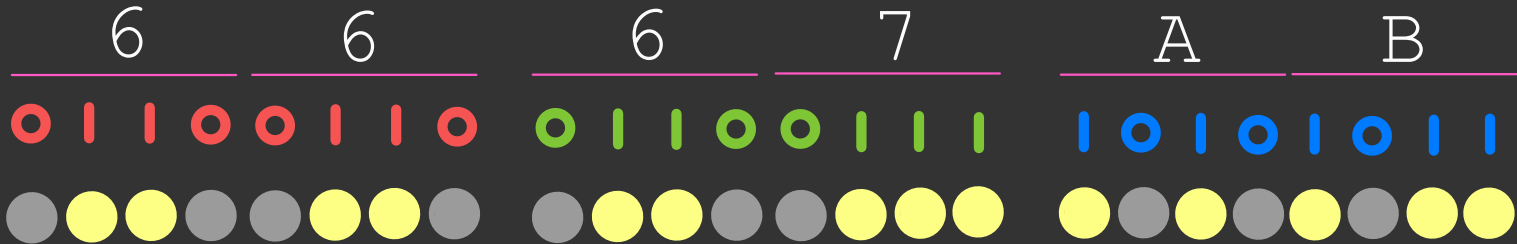
PANTONE
17-3938 TCX
Very Peri

Color Hex Code
#6667AB

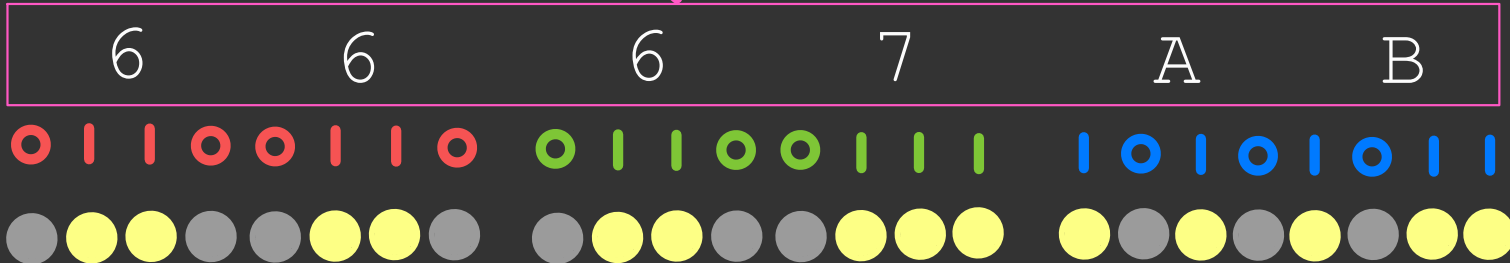


to save time,
we can choose
symbols to
represent the
16 different
nibbles

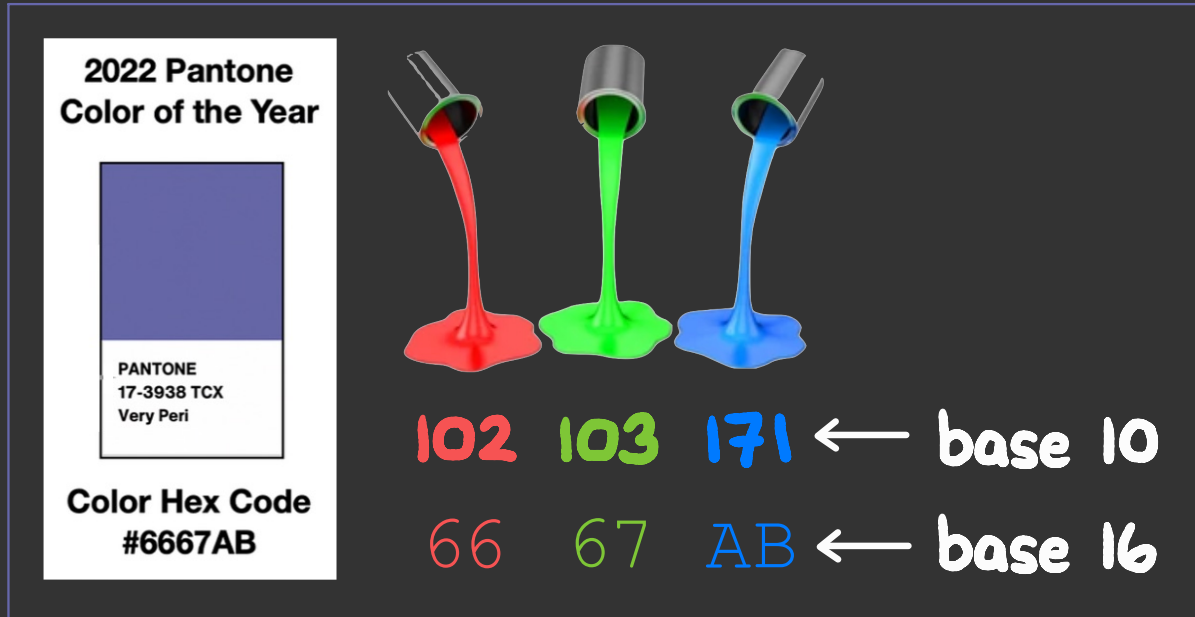
0 0 0 0	→	0	1 0 0 0	→	8
0 0 0 1	→	1	1 0 0 1	→	9
0 0 1 0	→	2	1 0 1 0	→	A
0 0 1 1	→	3	1 0 1 1	→	B
0 1 0 0	→	4	1 1 0 0	→	C
0 1 0 1	→	5	1 1 0 1	→	D
0 1 1 0	→	6	1 1 1 0	→	E
0 1 1 1	→	7	1 1 1 1	→	F



to save time,
we can choose
symbols to
represent the
16 different
nibbles



this is equivalent to representing each color channel with its base **16** representation



2022 Pantone Color of the Year

PANTONE 17-3938 TCX
Very Peri

Color Hex Code
#6667AB

102 103 171 ← base 10

66 67 AB ← base 16

base **16** is commonly called **hexadecimal**